$$\langle B| = \begin{pmatrix} \beta_1^* & \beta_2^* & \beta_3^* \end{pmatrix}.$$

The rule is again just matrix multiplication. With a slight abuse of notation,

$$\langle B|\mathbf{M} = \begin{pmatrix} \beta_1^* & \beta_2^* & \beta_3^* \end{pmatrix} \begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix}. \qquad (3.6)$$

3.1.3 Hermitian Conjugation

You might think that if **M**|*A*> = |*B*> then <*A*|**M** = <*B*|, but if you do you are wrong. The problem is complex conjugation. Even when *Z* is just a complex number, if *Z*|*A*> = |*B*>, it is not generally true that <*A*|*Z* = <*B*|. You have to complex-conjugate *Z* when going from kets to bras: <*A*|Z^* = <*B*|. Of course, if *Z* happens to be a real number, then complex conjugation has no effect—every real number is equal to its own complex conjugate.

What we need is a concept of complex conjugation for operators. Let's look at the equation **M**|*A*> = |*B*> in component notation,

$$\sum_i m_{ji}\alpha_i = \beta_j,$$

and form its complex conjugate,

$$\sum_i m_{ji}^*\alpha_i^* = \beta_j^*.$$

We would like to write this equation in matrix form, using bras instead of kets. In doing this, we have to remember that bra-vectors are represented by rows, not columns. For the result to work out correctly, we also need to